RACING TO THE SINK: WHY GATED ATTENTION STILL FORMS ATTENTION SINKS, AND WHAT DECIDES THE RACE

Aditya Kariwal (23MA10003), Manish Kundu (23MA10032) & Samyak Sheersh (22EC39026)
Group G28 — Foundations of Large Language Models (End-Term Project)

ABSTRACT

Ran-Milo (2026) proved that softmax attention *must* form an attention sink to implement a trigger-conditional "no-op", and pointed to explicit gating as the way to obtain sink-free models. We test this suggestion in the paper's own controlled setting and find that a sink is not only necessary for softmax, it is also *preferred* by training. Gated and softmax-off-by-one attention admit explicit zero-loss solutions without a sink, yet trained gated models keep full BOS sinks at every depth from 1 to 8 layers and for every gate initialisation. We show that this is the outcome of a *race* between two ways of switching a head off: the sink logit is a product of two learned factors and grows quadratically in time, while the gate logit grows only linearly. A sign-Adam analysis of the race makes three predictions, which we test in about 230 further runs. (i) The two routes share one fixed log-error budget: at convergence, sink logit plus gate closing is constant (8.8 ± 1.8, correlation -0.83). (ii) The outcome does not depend on the base learning rate (curves for a $10\times$ range coincide). (iii) The gate learning-rate multiplier needed to win, λ^*, grows like $\sqrt{d_h}$ with head width (fitted exponent 0.55); the law holds, but its constant is about $5\times$ larger than predicted. Two predictions fail, and we report them: plain SGD does not remove sinks, and in 4-layer models λ^* is almost width-independent. Controls make the race causal: speeding up the query/key weights restores every sink even when the gate is fast, while a $10\times$ faster gate alone removes all sinks in a 4-layer model. In small character-level language models trained for up to 20k steps no attention sink forms in any variant, so the language-model evidence is inconclusive.

1 INTRODUCTION

Large language models put a disproportionate share of attention on the first token, a phenomenon called the *attention sink* (Xiao et al., 2024). Sinks are linked to massive activations (Sun et al., 2024), complicate quantisation (Bondarenko et al., 2023), and matter for long-context inference. A common explanation is that a head sometimes has nothing useful to read; since softmax weights must sum to one, the head "parks" its attention on a token whose value contributes nothing (Miller, 2023; Gu et al., 2025; Guo et al., 2024).

Ran-Milo (2026) made this precise. On a *trigger-conditional* task, where a head must average the past only when a trigger token appears and output zero otherwise, they prove that any low-loss softmax model must put almost all attention on BOS at non-trigger positions, while ReLU attention solves the same task without a sink. Their practical conclusion (Sec. 5) is that removing sinks requires changing how "off" states are represented, "via non-normalized attention, explicit gating", and they state, citing Qiu et al. (2025), that sinks do not appear in gated attention.

This project asks whether that suggestion holds, and if not, what decides the outcome. If a model *can* switch a head off without a sink, does training use that ability? We study this in the exact setting of the original paper, where every quantity can be measured and controlled.

Contributions. Beyond reproducing the original results with our own implementation:

1. We give **explicit zero-loss constructions without a sink** for gated and off-by-one attention (Section 5), so for them a sink is *not* necessary.
2. We show that **training forms the sink anyway**. This holds across depths 1–8 and gate initialisations 0.02–0.98 (Section 6).
3. We show that the outcome is a **race**. Gradient probes and trajectories show a quadratic sink logit against a linear gate logit. Controls show the race is causal: accelerating W_Q, W_K restores every sink, while accelerating only the gate removes them all (Section 7).
4. We develop a **quantitative theory of the race** (Section 8) and test its predictions: a shared log-error budget, invariance to the base learning rate and a $\sqrt{d_h}$ law for the critical multiplier. All three hold, the last up to a constant factor of about 5. We also report the predictions that fail.
5. We evaluate **gate parameterisations** motivated by the theory (Section 9). We also train **small language models** (Section 10). No sink formed in them within our budget, so they do not yet test the race.

2 Background and Related Work

Attention sinks. Xiao et al. (2024) showed that keeping the first tokens in the KV cache is essential for streaming inference. Gu et al. (2025) studied when sinks emerge during pre-training, finding that they appear after effective optimisation, are weaker with small learning rates, and disappear with un-normalised sigmoid attention. Sun et al. (2024) linked sinks to massive activations, and Guo et al. (2024) explained them with active–dormant heads. Varre et al. (2026) show that gradient flow on value$\times$softmax models drives outputs to low-entropy solutions, a formal route to sinks.

Sinks are necessary for softmax. Ran-Milo (2026) prove that on trigger-conditional tasks softmax attention needs a sink (single-layer, Thm. 1; at least one layer of a deep model, Thm. 2), while ReLU attention admits a sink-free solution (Thm. 3). Their theorems extend to any attention whose weights are normalised ($\sum_k \alpha_{ik} \geq c > 0$) and monotone, as long as the head outputs the weighted average of its values. Off-by-one attention breaks normalisation, and ReLU and sigmoid attention break both conditions. Gated attention keeps the weights normalised and monotone but multiplies the average by a query-dependent factor, so it breaks the third assumption instead. None of the variants we add below is covered by the theorems.

Mechanisms that can switch off. Softmax-off-by-one adds a constant 1 to the softmax denominator, letting rows sum to less than one (Miller, 2023). Bondarenko et al. (2023) proposed clipped softmax and gated attention to help heads "do nothing". Qiu et al. (2025) add a head-specific sigmoid gate after attention and report strongly reduced sinks in LLMs trained on 3.5T tokens. Sigmoid attention (Ramapuram et al., 2025) removes normalisation altogether.

Closest work. Two very recent preprints are directly relevant. Rizwan et al. (2026) report that gating did *not* remove sinks in small, short-trained language models. Wang (2026) argue that a gate on the value pathway supplies an "abstention" primitive that softmax lacks. Our study explains when a gate is actually used for abstention and when training ignores it: the race of Sections 7 and 8. Saxe et al. (2014) analyse the sigmoidal learning curves of products of weights in deep linear networks, which is the mechanism behind the fast-growing sink route.

3 Setup

Task. We follow Ran-Milo (2026) and their released code exactly. A sequence has $L = 16$ tokens $x^{(0)}, \ldots, x^{(15)} \in \mathbb{R}^{16}$. Position 0 is BOS (indicator e_{bos}, zero content). One position $j \in \{2, \ldots, 15\}$ is the trigger (e_{trig}); all others are content tokens (e_{cont}). Every non-BOS token carries 13 i.i.d. $\mathcal{U}(-1, 1)$ content features. The target is the mean of $x^{(1)}, \ldots, x^{(j)}$ at the trigger and zero elsewhere (multi-layer models predict this on top of the residual stream). There is no positional encoding.

Models. Attention-only transformers with D layers of H heads of width d_h ($d_h = 16$ unless stated), causal masking, residual connections for $D > 1$, and no MLP or LayerNorm, as in the original paper.

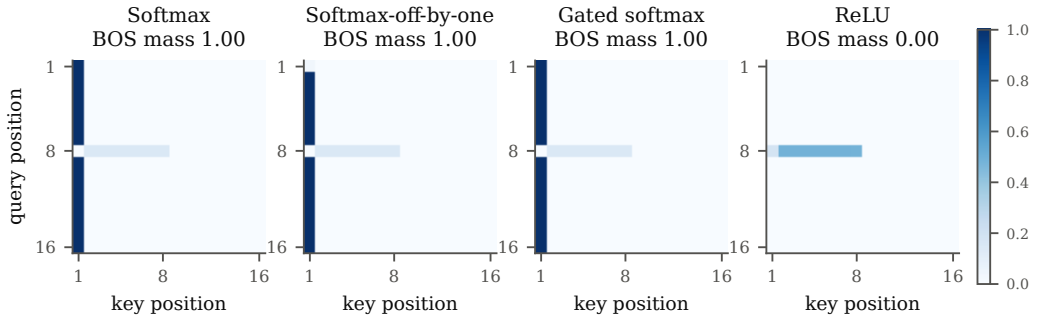


Figure 1: Mean attention of a trained single-layer, single-head model (trigger at position 8, 1000 test sequences, seed 0). Softmax, off-by-one and gated attention all park non-trigger rows on BOS (column 1); ReLU switches off instead.

With $s_{ik} = q_i^\top k_k$, the head output is $W_O\big(g_i \odot \sum_{k\le i} \alpha_{ik} v_k\big)$ with $g_i \equiv 1$ unless gated, and

$$\text{softmax: } \alpha_{ik} = \frac{e^{s_{ik}}}{\sum_{k'\le i} e^{s_{ik'}}}, \qquad \text{off-by-one: } \alpha_{ik} = \frac{e^{s_{ik}}}{1+\sum_{k'\le i} e^{s_{ik'}}},$$

$$\text{gated: softmax with } g_i = \sigma(W_G x_i + b_G), \qquad \text{ReLU: } \alpha_{ik} = \mathrm{ReLU}(s_{ik})/i,$$

$$\text{sigmoid: } \alpha_{ik} = \sigma(s_{ik} - \log L), \qquad \text{sigmoid-n: } \alpha_{ik} = \sigma(s_{ik})/i.$$

The gate is elementwise and head-specific and is computed from the query token, as in Qiu et al. (2025). "Sigmoid-n" uses the same $1/i$ length scaling as the paper's ReLU.

Training. Adam (Kingma & Ba, 2015) with $\beta = (0.9, 0.95)$, batch 128, learning rate 10^{-3} (1 layer, no residual) or 10^{-4} (residual models), weights $\mathcal{N}(0, 0.02^2)$, gate bias 0 unless stated. As in the reference code, training stops when the ℓ_∞ error on a batch falls below 5×10^{-3} (the paper's text says 10^{-2}) or after 50k steps (80k for 6–8 layers). λ denotes a learning-rate multiplier applied only to the gate parameters (W_G, b_G).

Metrics. Following the paper, we evaluate on 1000 sequences with the trigger at position 8 and measure, per head, the attention mass on BOS averaged over *non-trigger* rows ($i \ge 1$, $i \ne j$). A head is a *sink head* if this exceeds 0.6. For off-by-one we also report the *null-slot mass* $1 - \sum_k \alpha_{ik}$. For gated attention we report the gate value on non-trigger and trigger rows. Because the gate has d_h coordinates per head and not all of them carry signal, we also report an *importance-weighted* gate in which coordinate k of head h is weighted by $\|W_O[:, h, k]\| \, \|W_V[h, k, :]\|$. Results in the main text use 5 seeds (Secs. 4–6) or 3 seeds (Secs. 7–9).

4 Reproduction

Figure 1 and Table 1 reproduce the paper: softmax forms a perfect sink in every seed and ReLU never does. All 90 runs converge except un-normalised sigmoid, which cannot compute an average at 1–2 layers and only succeeds once a 4-layer residual stack can compensate. Sigmoid-n solves the task without a sink at every depth, consistent with Gu et al. (2025). At 4 layers, sinks appear in some heads but not all (37/80 heads over 5 seeds), matching the existential form of the paper's Theorem 2.

5 A sink is not necessary when the head can switch off

The proofs of Ran-Milo (2026) rely on the head's output being a normalised average of values. Gating multiplies that average by a query-dependent factor, and off-by-one lets the weights sum to less than one, so neither is covered. We show both admit sink-free solutions, in the spirit of the paper's ReLU construction (Thm. 3). Let $W_K = W_V = W_O = I$ and write the score as $s_{ik} = x_i^\top B\, x_k$.

Proposition 1 (Gated attention). *Let* $B = -c\,(e_{trig} + e_{cont})e_{bos}^\top$ *and* $g_i = \sigma\big(a\,(x_i[trig] - x_i[cont] - x_i[bos])\big)\mathbf{1}$. *Then every non-*BOS *query gives score* $-c$ *to* BOS *and* 0 *to all other keys, so* $\alpha_{i,\mathrm{BOS}} =$

Table 1: Reproduction and variants (5 seeds each). "Max BOS": largest per-head BOS mass on non-trigger rows, mean ± std over seeds. "Sink heads": heads with BOS mass > 0.6, summed over seeds. DLHH: D layers, H heads.

Model	Attention	Conv.	Steps	Max BOS	Sink heads	Gate off / trig.
1L1H	softmax	5/5	905	$1.00 \pm .00$	5/5	–
	off-by-one	5/5	911	$1.00 \pm .00$	5/5	–
	gated	5/5	1002	$1.00 \pm .00$	5/5	0.51 / 0.79
	ReLU	5/5	692	$0.00 \pm .00$	0/5	–
	sigmoid	0/5	–	–	–	–
	sigmoid-n	5/5	831	$0.18 \pm .03$	0/5	–
2L2H	softmax	5/5	10509	$1.00 \pm .00$	20/20	–
	off-by-one	5/5	8921	$1.00 \pm .00$	20/20	–
	gated	5/5	9794	$1.00 \pm .00$	20/20	0.45 / 0.66
	ReLU / sigmoid-n	5/5	–	0.00 / 0.13	0/20	–
4L4H	softmax	5/5	22379	$1.00 \pm .00$	37/80	–
	off-by-one	5/5	18155	$0.89 \pm .10$	40/80	–
	gated	5/5	16816	$0.99 \pm .01$	38/80	0.47 / 0.59
	ReLU / sigmoid-n	5/5	–	0.00 / 0.16	0/80	–

Table 2: Explicit sink-free solutions (Propositions 1 and 2) evaluated on 2000 sequences. The training stopping threshold is $\ell_\infty < 5 \times 10^{-3}$.

	Gated			Softmax-off-by-one			
Scale	MSE	ℓ_∞	BOS mass	MSE	ℓ_∞	BOS mass	Row mass
2	1.5×10^{-3}	0.174	2.9×10^{-2}	1.8×10^{-2}	0.625	6.6×10^{-2}	0.51
5	5.2×10^{-6}	0.010	1.5×10^{-3}	2.3×10^{-4}	0.086	6.4×10^{-3}	0.056
10	2.4×10^{-10}	10^{-4}	1.0×10^{-5}	1.2×10^{-8}	6×10^{-4}	4.5×10^{-5}	4×10^{-4}
20	≈ 0	≈ 0	4.6×10^{-10}	≈ 0	≈ 0	2.1×10^{-9}	2×10^{-8}

$e^{-c}/(e^{-c}+i) \le e^{-c}$ *at* every *row. The gate equals* $\sigma(a)$ *at the trigger and* $\sigma(-a)$ *elsewhere. Since inputs are bounded, the maximum error over all inputs is* $O(e^{-a}+e^{-c})$*. Hence for any* $\varepsilon > 0$ *there are weights with loss* $\le \varepsilon$ *and* BOS *mass* $\le \varepsilon$ *at all non-trigger rows.*

Proposition 2 (Softmax-off-by-one)**.** *Let* $B = m\big[e_{trig}(e_{trig} + e_{cont} - e_{bos})^\top - (e_{bos} + e_{cont})(e_{bos} + e_{trig} + e_{cont})^\top\big]$*. The trigger query scores* $+m$ *on every non-*BOS *key and* $-m$ *on* BOS*, so its weights tend to the uniform average over positions* $1..j$*. Every other query scores* $-m$ *on all keys, so its total mass is* $\le i\,e^{-m}$ *and it outputs* ≈ 0 *through the null slot. Again the error is* $O(Le^{-m})$ *and the* BOS *mass vanishes.*

Table 2 evaluates both constructions numerically on 2000 random sequences. Already at scale 10, both reach an error far below the training threshold with essentially zero BOS attention. Notably, the gated solution uses a *query-independent* attention pattern and puts all of the conditional behaviour into the gate.

6 . . . BUT TRAINING LEARNS THE SINK ANYWAY

Gated attention sinks like softmax. Despite Proposition 1, every trained gated model in Table 1 parks non-trigger rows on BOS, with as many sink heads as softmax (38/80 vs. 37/80 at 4 layers). The gate barely moves from its initial value of 0.5: it ends at about 0.45–0.5 on non-trigger rows and 0.6–0.8 on the trigger row. This is a mild modulation, not a switch. The sink does the switching.

Gate initialisation does not help. One might expect a gate that starts nearly closed to be used as a switch. It is not (Figure 8 in Appendix B). With the gate initialised at $\sigma(-4) = 0.02$ or $\sigma(-2) = 0.12$, *all* 80/80 heads become sinks (vs. 38/80 at 0.5). In every setting the final gate stays close to its initial value (e.g. $0.02 \to 0.02/0.05$ off/trigger).

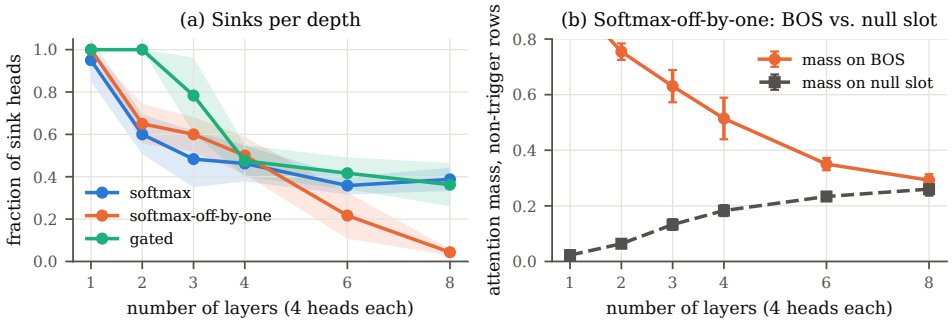


Figure 2: Depth sweep (4 heads per layer, 5 seeds; band/bars: ± 1 std). (a) Fraction of sink heads. (b) For off-by-one, attention mass on BOS vs. the null slot, averaged over all heads.

Table 3: Training without a BOS token (5 seeds per cell). "Mass": mean total attention $\sum_k \alpha_{ik}$ on non-trigger rows (1 for softmax and gated by construction).

Attention	1 layer, 1 head	2 layers, 2 heads	4 layers, 4 heads
softmax	0/5 ($\ell_\infty = 0.74$)	0/5 (0.70)	0/5 (0.08)
off-by-one	5/5, mass 0.00	5/5, mass 0.42	5/5, mass 0.57
gated	5/5, gate 0.02 / 0.62	5/5, gate 0.28 / 0.63	5/5, gate 0.41 / 0.64
ReLU, sigmoid-n	5/5	5/5	5/5

Depth. Figure 2 sweeps depth from 1 to 8 layers with 4 heads per layer. Softmax and gated attention keep sinks at every depth. A declining fraction of heads are sinks as depth grows, but the strongest head always has BOS mass ≥ 0.95. Off-by-one behaves differently. At 1–2 layers it is indistinguishable from softmax, but with depth its sinks fade (7/160 sink heads at 8 layers) and mass moves into the null slot.

Removing the BOS token. If training prefers the sink because a convenient anchor exists, it should use the off switch once the anchor is gone. We drop BOS, so every position is an ordinary content token. Table 3 confirms the prediction: off-by-one and gated attention always solve the task, and the gated model now uses a genuine switch (gate 0.02 off vs. 0.62 on at one layer). Softmax cannot solve the task at any depth (0/15 runs) and forms no substitute anchor (no head puts more than 0.21 on any single key). We also repeat the experiment with *learned positional embeddings* added to the query and key inputs, which lets a head single out a position. Softmax still fails and still forms no anchor (maximum 0.20 on any key, 3 seeds at 1 and 2 layers). The gated model still solves the task with gate 0.01. This failure is what the theorem predicts rather than a counter-example to it: an anchor is only useful if its value is constant, and without BOS every token's value carries random content. The expectation of Ran-Milo (2026, Sec. 5) that a suppressed sink "reappears elsewhere" therefore needs a token with a constant value, such as a delimiter or a repeated special token, to be tested.

7 Why the sink wins: a race between two routes

On a non-trigger row a gated head can switch off in two ways. It can put its attention on BOS (whose value it can set to zero), or it can close its gate. We call these the sink route and the gate route.

Gradient probes. We add two zero-valued probe scalars per head, active only on non-trigger rows: p_{sink} is added to the BOS attention logit and p_{gate} to the gate pre-activation. They leave the forward pass unchanged, and their gradients show which route the loss pulls on. Figure 3(a) corrects an earlier version of this analysis. Because p_{gate} acts on all $d_h = 16$ gate coordinates at once, its raw gradient is a *sum* over 16 coordinates and looks larger than the sink pull (peak 6.2×10^{-4} vs. 3.9×10^{-4}). Per coordinate, the gate pull is only 0.1–0.4$\times$ the sink pull. Under Adam the size of a gradient does not set the size of an update in any case. The question is therefore which route *moves* faster, not which one the loss prefers.

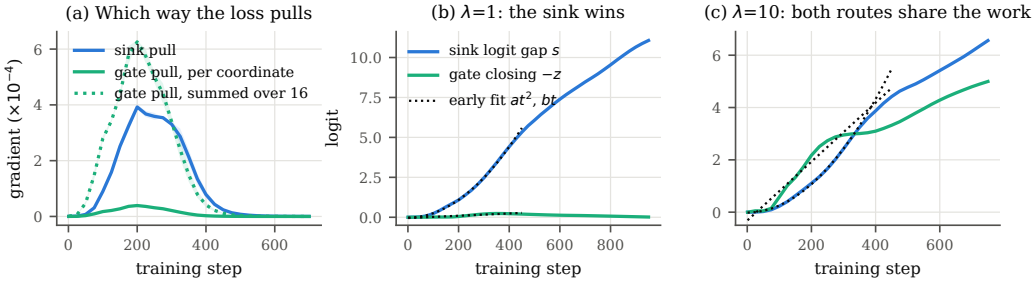


Figure 3: Single-layer gated model, 3 seeds. (a) Probe gradients: the gate pull is large only when summed over its 16 coordinates (dotted); per coordinate it is far smaller than the sink pull. (b, c) The two routes' logits on non-trigger rows: the sink grows like t^2, the gate like t (dotted: early-phase fits).

Table 4: Controls (4 layers $\times$ 4 heads, $d_h = 16$, 3 seeds, all converged). Gate values are importance-weighted, on non-trigger / trigger rows.

Configuration	Sink heads	Mean BOS	Gate off / trig.	Steps
baseline ($\lambda = 1$)	23/48	0.57	0.47 / 0.58	18.4k
gate lr $\times 10$ ($\lambda = 10$)	**0/48**	0.20	0.16 / 0.71	3.3k
gate lr $\times 10$ and W_Q, W_K lr $\times 10$	**48/48**	1.00	0.74 / 0.91	3.4k
all lr $\times 10$	33/48	0.74	0.43 / 0.58	3.4k
scalar gate per head, $\lambda = 1$	28/48	0.63	0.44 / 0.60	12.8k
scalar gate per head, $\lambda = 10$	0/48	0.32	0.04 / 0.74	4.2k

Trajectories. Figure 3(b,c) tracks the BOS logit gap s (the BOS logit minus the mean logit of the other keys) and the gate closing $-z$ (minus the mean gate pre-activation) on non-trigger rows. At $\lambda = 1$ the sink logit rises quadratically, $s \approx at^2$, while the gate barely moves; the run ends with $s \approx 11.5$, $-z \approx 0$. At $\lambda = 10$ the gate closes linearly, $-z \approx bt$, and gets an early lead. The quadratic sink then catches up, and both routes end with a share of the work ($s \approx 7$, $-z \approx 5$, BOS mass 0.99, gate 0.01).

Controls. Table 4 intervenes on the speed of each route in the 4-layer model. A $10\times$ gate learning rate removes every sink (0/48 heads). Speeding up *also* the query and key weights (W_Q, W_K, the factors of the sink logit) by $10\times$ turns every head into a sink (48/48). Raising *all* learning rates by $10\times$ trains just as fast but keeps 33/48 sinks. So the effect is the relative speed of the two routes, not the amount of learning rate. It also shows that the speed-up in training time is a property of a larger learning rate rather than of the gate. A single scalar gate per head behaves like the elementwise gate.

8 A QUANTITATIVE THEORY OF THE RACE

Minimal model. Consider one non-trigger row of a gated head. Ignoring the other heads, its output is $g_i \odot (1 - \alpha_{i,\text{BOS}})\bar{v}_i$, where $\bar{v}_i$ is the attention-weighted mean of the non-BOS values. With i non-BOS keys, $1 - \alpha_{i,\text{BOS}} \approx i\, e^{-s}$ and $\sigma(z) \approx e^z$ when the routes are well under way. The row's error is therefore

$$\log \text{err} \;\approx\; \log(C\, i) \;-\; s \;-\; (-z), \qquad C = \|\bar{v}\|_\infty. \tag{1}$$

Training stops when err $< \varepsilon$, so at that moment

$$s^* + (-z^*) \;\approx\; T' := \log(C\, i/\varepsilon). \tag{2}$$

The two routes *share one budget*. Whatever the sink has done when training stops, the gate does not have to do, and the other way round.

Speeds under Adam. With a small learning rate η, Adam moves each weight by about η per step as long as the sign of its gradient is stable. The sink logit is a sum of d_h products, $s = \sum_{r \le d_h} W_Q[r, \text{cont}]\, W_K[r, \text{bos}]$. Each factor grows linearly, so $s(t) \approx d_h(\eta t)^2$. The gate logit

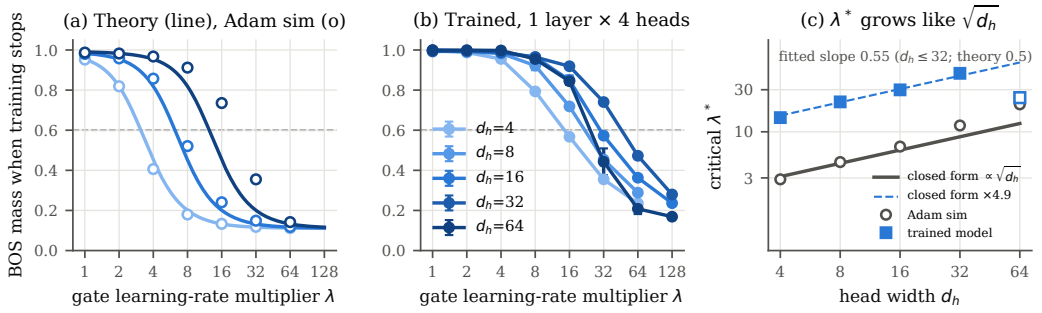


Figure 4: The critical gate multiplier. (a) Minimal model: closed form (lines) and Adam simulation (circles). (b) Trained 1-layer, 4-head models (3 seeds): BOS mass vs. λ per head width; dashed: sink threshold. (c) λ^* (where the BOS mass crosses 0.6) vs. d_h.

$z = W_G[\cdot, \text{cont}] + b_G$ is linear in two weights whose learning rate is $\lambda\eta$, so $-z(t) \approx 2\lambda\eta t$. Writing $u = \eta t$, training stops at $d_h u^2 + 2\lambda u = T'$. The head ends as a sink ($\alpha_{\text{BOS}} > a_0 = 0.6$) iff $s^* = d_h u^{*2} > S_0 := \log(i\, a_0/(1 - a_0))$. Solving for the multiplier at which this switches,

$$\lambda^*(d_h) \;=\; \frac{T' - S_0}{2\sqrt{S_0}}\,\sqrt{d_h}. \tag{3}$$

This gives three testable predictions. **(P1)** the routes share a constant budget, Equation (2); **(P2)** η cancels, so the outcome does not depend on the base learning rate; **(P3)** λ^* grows like $\sqrt{d_h}$. With $i = 8$, $\varepsilon = 5 \times 10^{-3}$ and $C = 1$ we get $T' = 7.4$, $S_0 = 2.5$ and $\lambda^* \approx 1.55\sqrt{d_h}$. A direct simulation of the minimal model with real Adam (not the sign approximation; `race_theory.py`) agrees with the closed form (Figure 4a). Under SGD the analysis changes: the sink's gradient is proportional to the other factor, which starts at $O(0.02)$. The minimal model therefore predicts **(P4)**: with SGD the gate always wins.

Tests. We sweep $d_h \in \{4, \ldots, 64\}$ and $\lambda \in \{1, \ldots, 128\}$ in the 1-layer, 4-head residual model (3 seeds per cell; Figure 4b,c).

- **P1 holds.** Across all 1-layer runs, $s^* + (-z^*) = 8.8 \pm 1.8$ per head, and the two are strongly anti-correlated ($r = -0.83$; Figure 5b). The budget is larger than our rough constant ($T' \approx 7.4$) because the true C and i differ from the values we assumed.
- **P2 holds.** At base learning rates 3×10^{-5}, 10^{-4} and 3×10^{-4} the BOS mass vs. λ curves are indistinguishable (Figure 5a). Training takes $10\times$ more steps at the smallest rate, but the outcome is the same.
- **P3 holds up to a constant.** λ^* is 14.5, 21.8, 29.9 and 46.0 at $d_h = 4, 8, 16, 32$. The fitted log-log slope is 0.55 (theory 0.5), and every value is $4.8 \pm 0.2\times$ the closed form. The exception is $d_h = 64$ ($\lambda^* = 24$), where the head is wider than the model ($d = 16$): $W_O W_V$ has rank at most 16, so only a quarter of the gate coordinates matter. The gate route is slower than the two-weight idealisation assumes, because its gradient sign is noisy: it is driven by random content and opposed on the trigger row.
- **P4 fails.** With SGD (momentum 0.9, lr 30; 3 seeds), gated models still form sinks in 5 of 12 heads. In the full model the gate's gradient is itself proportional to the small value and output weights, a coupling the minimal model leaves out. So both routes start slowly under SGD.

In deeper models the race is less width-dependent: at 4 layers, $\lambda = 10$ is enough at every width tested (Section 9). This is plausibly because several heads can cancel each other's leftover output, which gives the network a third way to switch off.

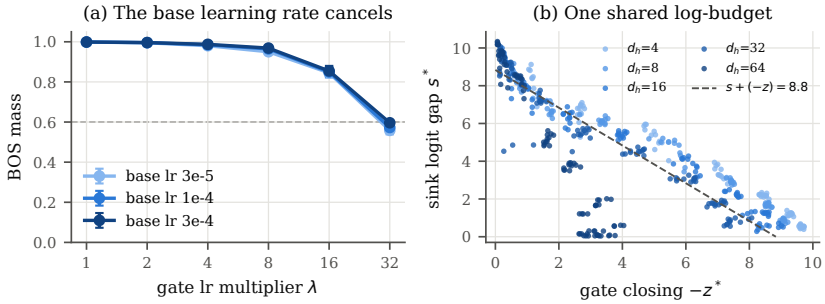


Figure 5: (a) $d_h = 16$: changing the base learning rate $10\times$ does not change the race (3 seeds per point). (b) One point per head of every 1-layer run: sink logit gap vs. gate closing at the stopping point. They trade off along $s + (-z) =$const, as predicted by Equation (2).

9 Changing the Race: Gate Parameterisations

If the outcome is a race, the gate can be made to win by changing its parameterisation instead of hand-tuning a learning rate. We compare four gates in the 4-layer model at $d_h \in \{4, 16, 64\}$ and depth 1, 4 and 8 (Figure 6, 3 seeds each):

- the **linear gate** of Qiu et al. (2025) at $\lambda = 1$ and $\lambda = 10$;
- a **race-matched gate** $g = \sigma(\kappa(W_G x + b))$ with $\kappa = 2.5\sqrt{d_h}$ and W_G initialised at $0.02/\kappa$. It computes the same function as the linear gate at initialisation and under Adam is exactly $\lambda = \kappa$. The constant 2.5 matches the validated $\lambda = 10$ at $d_h = 16$, and the $\sqrt{d_h}$ rule comes from Equation (3);
- a **bilinear gate** $z = A_h(B_h x) + b$, a product of two learned factors like the sink. The theory predicts that a symmetric race splits the budget evenly, giving partial sinks independent of d_h.

Results.

- The race-matched gate is sink-free at $d_h = 16$ and 64 (0/48 each) and at 8 layers (0/96). It is also the fastest to converge (1.7k steps at $d_h = 64$ vs. 9.8k at $\lambda = 1$).
- At $d_h = 4$ it fails (23/48 sinks with $\kappa = 5$), while the fixed $\lambda = 10$ succeeds at all three widths. In the 4-layer model the $\sqrt{d_h}$ rule therefore scales the gate down too much for narrow heads.
- The bilinear gate gives partial sinks, as predicted (mean BOS 0.32–0.66). It is not a fix.

The practical recommendation is a gate learning-rate multiplier of order 10 at 4–8 layers. At one layer, λ^* grows with width and a larger multiplier is needed (Figure 4c). Table 5 in Appendix B lists the full λ sweep.

10 A Small Language Model

Does the race carry over to a real language model? We train character-level GPTs on TinyStories (`lm.py`): 6 layers, $d = 384$, 6 heads of width 64, pre-LayerNorm with MLPs, learned positional embeddings, context 256, AdamW ($\beta = (0.9, 0.95)$, weight decay 0.1, cosine schedule), batch 64, mixed precision on Kaggle T4s, about 11M parameters. Every sequence starts with a BOS token. We compare softmax, off-by-one, the linear gate at $\lambda = 1$ and $\lambda = 10$, and the race-matched gate ($\kappa = 2.5\sqrt{64} = 20$).

Run 1 (first 255 characters of each story, 8k steps, learning rate 10^{-3}, 2 seeds; Figure 7). After 131M characters *no* variant has an attention sink. Mean attention on BOS is 0.02 for all of them, the uniform level, and the strongest BOS head reaches only 0.15. This agrees with Gu et al. (2025),

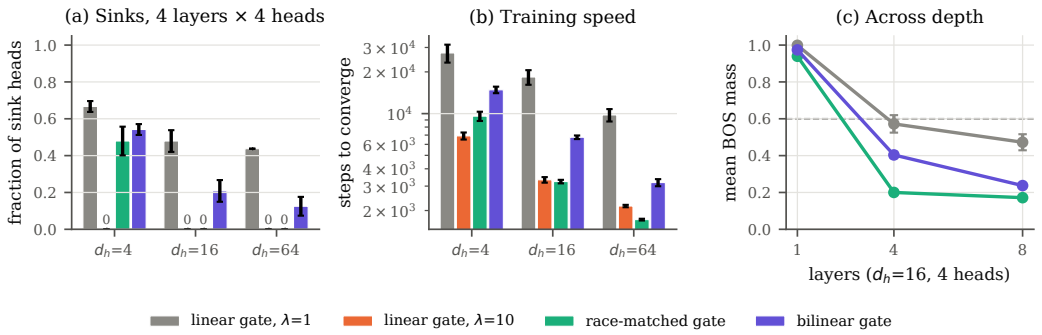


Figure 6: Gate parameterisations (3 seeds). (a) Fraction of sink heads at 4 layers × 4 heads, by head width. (b) Steps to converge. (c) Mean BOS mass vs. depth at $d_h = 16$.

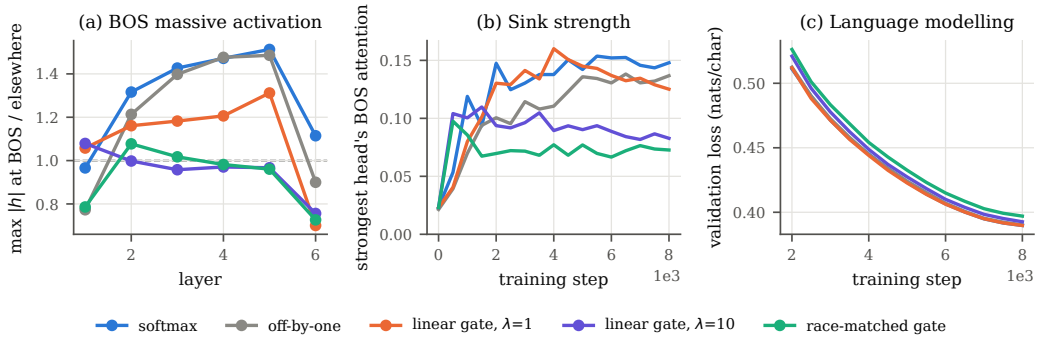


Figure 7: Character-level GPT on TinyStories, run 1 (2 seeds, 8k steps). (a) Largest hidden-state entry at BOS divided by the largest elsewhere, per layer, at the end of training: softmax and off-by-one build a BOS outlier, fast gates do not. (b) Attention of the most BOS-focused head during training. (c) Validation loss.

who find that sinks emerge only after sufficient training. What we do see is the *precursor* of a sink, a massive activation at the BOS position. In softmax models the largest hidden-state entry at BOS is $1.43\times$ the largest anywhere else (layers 2–5), and off-by-one is close ($1.39\times$). Gating weakens this outlier in the order the race predicts: $1.22\times$ at $\lambda = 1$, and none at all at $\lambda = 10$ ($0.97\times$) or with the race-matched gate ($1.01\times$). The strongest BOS head follows the same order (0.15, 0.14, 0.13, 0.08, 0.07). Faster gates cost a little language-modelling loss (0.390 for softmax and $\lambda = 1$ vs. 0.393 and 0.397 nats/char) at this short training length.

Run 2 (packed chunks from random offsets, as in standard pre-training; 20k steps, learning rate 2×10^{-3}, 1 seed; Figure 11 in Appendix B) was designed to make sinks form, but still has none: the strongest BOS head reaches 0.33 (softmax), 0.27 (gate, $\lambda = 10$), 0.18 ($\lambda = 1$) and 0.11 (off-by-one). The BOS activation outlier is now similar in all variants (1.26–$1.33\times$), so the ordering seen in run 1 does not hold under packed training. Many heads in every variant put almost all of their attention on a single key (up to 0.99), but we did not record which key, and in a character-level model such heads may simply be previous-character heads. Gating slightly *improves* the loss here (0.506 at $\lambda = 1$ vs. 0.520 for softmax, one seed), in line with Qiu et al. (2025). Overall, at this scale and training length the language model does not test the race. Whether a fast gate removes sinks once they form in real language models remains open.

11 Discussion

Implications for the original paper. Ran-Milo (2026) show what softmax *needs*; we show what training *prefers*. Adding an off switch makes the sink unnecessary, but it does not by itself make a

model sink-free, because gradient descent can reach the sink first. Whether it does is decided by a race whose structure we can predict: a shared log-budget and invariance to the base learning rate. Its threshold depends on head width and depth.

Relation to gated LLMs. Qiu et al. (2025) observe far fewer sinks with gating at scale, while Rizwan et al. (2026) do not at small scale. The race offers a common explanation. Whether the gate wins depends on the relative speed of the two routes, which depends on head width, depth, normalisation layers, initialisation, the per-parameter learning rate and training length, and all of these differ between the two studies. Our results suggest checking gate statistics early in training and raising the gate's learning rate when sink-free behaviour is desired.

Limitations.

- Most experiments use the synthetic task and attention-only models of the original paper: no MLPs, LayerNorm or positional encoding, sequence length 16 and width 16.
- The constructions are existence results, not a characterisation of all solutions.
- The race theory is a first-order sign-Adam argument. It predicts the budget and the learning-rate invariance well, but it gets the size of λ^* wrong by a factor of 4–5 and fails for SGD.
- The language models are small and no sink formed in them within 20k steps (Section 10), so the race is established on the synthetic task only.
- Some numbers differ slightly between CPU and GPU runs of the same seed. All the controls in Tables 4 and 5 are compared within one device.

12 Conclusion

Attention sinks are necessary for softmax, and they are also *preferred* by training whenever a stable anchor exists, even in architectures that could do without them. In a controlled setting we showed that gated and off-by-one attention admit sink-free solutions, yet gradient descent reliably builds sinks in them because the bilinear sink outruns the linear gate. We turned this into a quantitative race with a conserved log-budget and a learning-rate-invariant threshold, confirmed its main predictions, and identified where it breaks down. Removing the anchor, adding depth, or giving the gate a large enough learning-rate multiplier shifts the race towards the gate.

Code. All code, configurations and raw results are included with this submission (`https://github.com/Adityakariwal/fllm-final-project`). Every number and figure can be regenerated with `reproduce.sh`, and `src/summarize.py` recomputes every table entry from the raw results.

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A Implementation details

Code. We re-implemented the task and models from scratch in PyTorch, following the released reference code of Ran-Milo (2026) for data generation, initialisation, optimiser and stopping rule. Differences from the reference: batched data generation on the device, additional attention variants, head width d_h, gate parameterisations, probes and tracking. The reference stopping threshold (5×10^{-3}, from the code) is used throughout.

Hyper-parameters. $d = 16$, $L = 16$, batch 128, Adam $(0.9, 0.95)$, no weight decay, initialisation $\mathcal{N}(0, 0.02^2)$ for all weights, gate bias 0 unless stated. Learning rate 10^{-3} for single-layer non-residual models (Table 1 top block, gate-LR 1-layer sweep, probes) and 10^{-4} otherwise. The depth sweep and the race sweeps of Section 8 use residual models with 4 heads for all depths, including $D = 1$. Maximum steps: 50k (80k for 6 and 8 layers); the no-BOS 1-layer sweep was capped at 20k steps. SGD runs use momentum 0.9 and learning rate 30. This is the only rate in $\{1, 10, 30\}$ that converges within 6k steps; the loss is a mean over $128 \times 16 \times 16$ entries, so gradients are small.

No-BOS variant. Position 0 becomes a content token, the trigger is sampled from $\{1, \ldots, 15\}$, and the target averages positions $0..j$. ReLU and sigmoid-n divide by the number of non-BOS keys, which is $i+1$ without BOS instead of i. The positional-embedding variant adds a learned $\mathcal{N}(0, 0.02^2)$ embedding to the query and key inputs only, so the target is unchanged.

Probes. Every 25 steps we draw 512 fresh sequences, add zero scalars p_{sink} (to the BOS logit) and p_{gate} (to the gate pre-activation, all coordinates) on non-trigger rows of each head, and record $-\partial L/\partial p_{\text{sink}}$ and $\partial L/\partial p_{\text{gate}}$ (the latter both summed over and divided by the d_h coordinates). Probes are removed before the optimisation step, so training is unaffected.

Compute. Single-layer and shallow sweeps ran on a laptop CPU. Deeper sweeps of the original study ran on Kaggle (2× NVIDIA T4). The race sweeps of Sections 7 to 9 ran on 11 CPU cores, and the language models on Kaggle T4s. In total we trained 558 models (312 in the original study, 232 in the race sweeps, 14 language models), plus 6 probe runs.

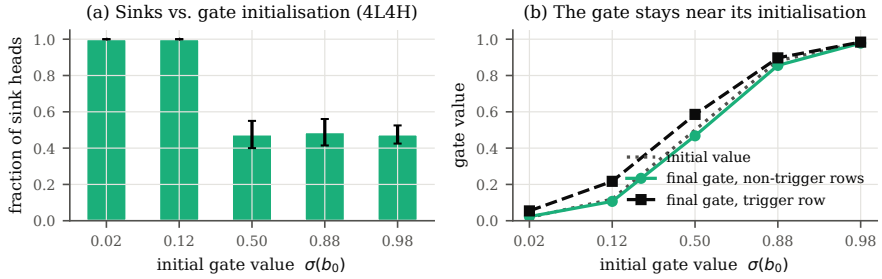


Figure 8: Gated attention, 4 layers × 4 heads, 5 seeds per bar. (a) A nearly closed initial gate turns every head into a sink. (b) The trained gate stays near its initialisation (dotted) instead of becoming an on/off switch.

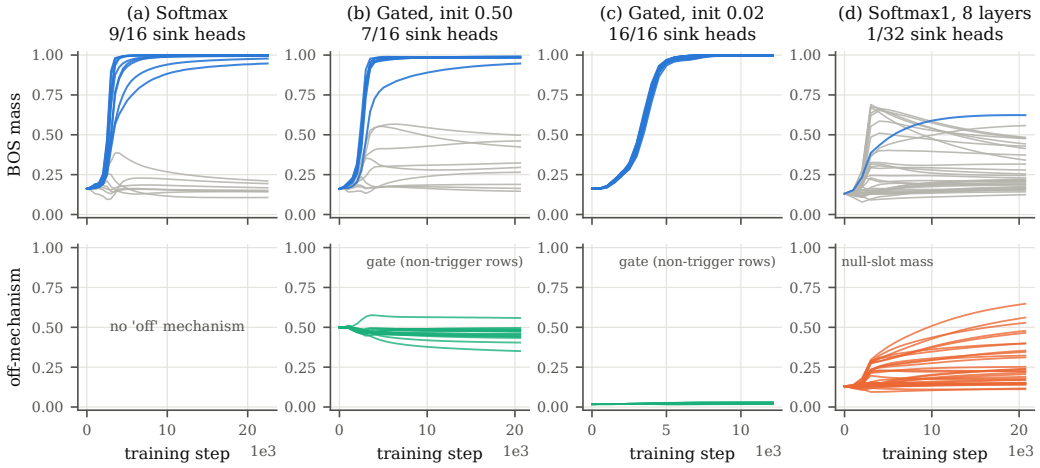


Figure 9: Training timelines (seed 0, 4 layers × 4 heads unless noted). Top: BOS mass per head (blue: heads that end as sinks). Bottom: the head's off mechanism, i.e. gate value (green) or null-slot mass (orange). Sinks appear abruptly (an S-shaped rise between roughly 2k and 5k steps) and never reverse in softmax and gated models. In deep off-by-one models sinks form first and are then slowly replaced by null-slot mass.

B Additional results

Sink heads by depth (5 seeds, 4 heads per layer, sink head: BOS mass > 0.6):

Layers	1	2	3	4	6	8
softmax	19/20	24/40	29/60	37/80	43/120	62/160
off-by-one	20/20	26/40	36/60	40/80	26/120	7/160
null-slot share in sink heads	0.02	0.02	0.11	0.19	0.32	0.33
gated	20/20	40/40	47/60	38/80	50/120	58/160
gate off / trig.	.42/.66	.43/.62	.45/.60	.47/.59	.47/.57	.48/.56

Gate initialisation (4 layers × 4 heads, 5 seeds): initial gate 0.02: 80/80 sink heads, final gate $0.02/0.05$; 0.12: 80/80, $0.11/0.22$; 0.50: 38/80, $0.47/0.59$; 0.88: 39/80, $0.86/0.90$; 0.98: 38/80, $0.98/0.98$ (off/trigger).

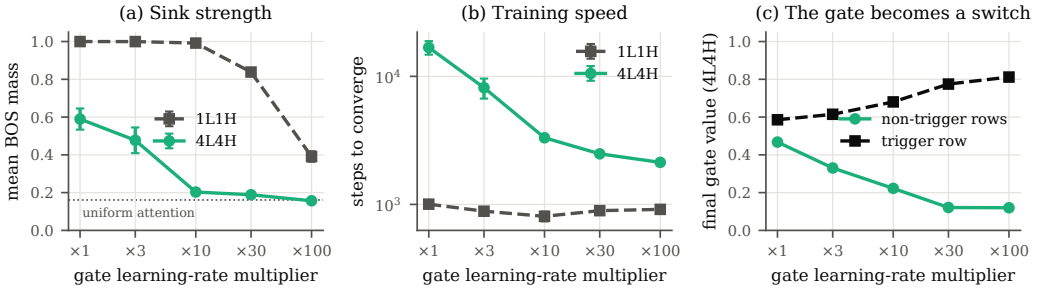


Figure 10: Gate learning-rate multiplier in the original configurations (5 seeds per point). (a) Sink strength. (b) Steps to converge. (c) Final gate values at 4 layers (mean over coordinates).

Table 5: Gate learning-rate multiplier λ in the original configurations (gated attention, 5 seeds each, all converged). At 1 layer, $\lambda = 10$ closes the gate *and* keeps the sink: the two routes share the budget (Section 8). At 4 layers the mean gate stays at 0.12–0.33 on non-trigger rows. Even there the no-op is not achieved by the gate alone; partial gating combines with heads cancelling each other.

	1 layer, 1 head			4 layers, 4 heads		
λ	BOS mass	Gate off/trig.	Steps	Sink heads	Gate off/trig.	Steps
1	1.00	0.50 / 0.78	1004	38/80	0.47 / 0.59	16816
3	1.00	0.33 / 0.93	885	25/80	0.33 / 0.61	8161
10	0.99	0.01 / 0.99	807	0/80	0.22 / 0.68	3315
30	0.84	0.00 / 1.00	894	0/80	0.12 / 0.77	2484
100	0.39	0.00 / 1.00	915	0/80	0.12 / 0.81	2130

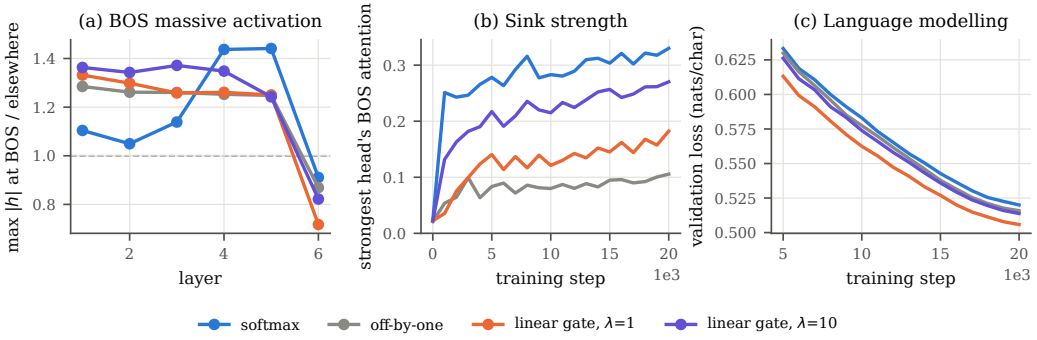


Figure 11: Language model, run 2 (packed random-offset chunks, 20k steps, 1 seed). (a) BOS activation outlier per layer, (b) attention of the most BOS-focused head, (c) validation loss. No sink forms, and the variants no longer differ in the BOS outlier.